

pyhydra

Data sources

Rainfall and climate: ERA5, GPM, PERSIANN-CCS, AEMET, CMIP6
Discharge and soils: GloFAS, GRDC, USGS, SoilGrids

Climate and statistics

Extremes and regional frequency analysis
Copulas and stochastic generation
Bias correction and hybrid downscaling

Model integration

Hydrology: HEC-HMS and SWAT+
Hydraulics: SFINCS and HEC-RAS
Ensemble execution and sensitivity analysis

Interoperable exchange layer `pandas.DataFrame` `xarray.Dataset` `GeoTIFF` `GeoDataFrame`

pyhydra scientific core modular Python APIs | lazy optional dependencies | unit-tested components

HYDRA execution and access layer JupyterLab | FastAPI | Astro/nginx | Docker Compose | Azure Container Apps